



Society of Petroleum Engineers

SPE-229947-MS

IRIS – Transforming Well Interventions Through Innovative Methodology to Enhance Oil Production in Electrical Submersible Pump Completions

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229947-MS](https://doi.org/10.2118/229947-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

In brown fields affected by scale, particularly those with Electric Submersible Pump (ESP) completions, traditional remediation often requires costly and complex rig-based workovers. To address this challenge, the Integrated Rigless Stimulation (IRIS) methodology was developed and successfully implemented in a mature field located in the Amazon Region of Ecuador. This novel approach enables chemical stimulation through a sliding sleeve using a Coiled Tubing Unit (CTU), without retrieving the ESP or mobilizing a rig.

The IRIS workflow consists of three stages: (1) Initial Recognition, where real-time surveillance is used to detect production anomalies and confirm whether they are related to a drop in the well's Productivity Index (PI) or due to operational changes, based on ESP parameter trends and nodal analysis. (2) Analysis, which follows a rigorous diagnostic process to validate the presence of scale near the perforated interval and reduce uncertainty before intervention. This stage integrates ESP trends, production and intervention history, fluid, completion and reservoir data, chemical treatment indicators, and offsets well behavior to build a consistent technical case. The evaluation is guided by engineering judgment and aims to confirm the root cause of flow restriction while confidently selecting stimulation candidates. (3) Implementation, where the stimulation design is customized based on the well diagnostic profile, including mechanical condition, scale severity, and reservoir properties. The rigless operation is then executed using a CTU, displacing the chemical treatment through the production string. The fluid is precisely placed at the sand face under controlled pressure conditions, enabling direct contact with the scale-affected areas and the wellbore while maintaining ESP integrity.

The IRIS methodology was applied to 18 wells in 2024, achieving PI improvements of 200–300%, representing a total incremental gain of 1,060 BOPD year average, and more than USD 25 million of revenue in combined production and cost savings. The intervention time was reduced from 15 days to 2 days per well, with a 75% decrease in downtime, an 85% reduction in operating cost, and a 75% cut in carbon dioxide (CO₂) emissions.

The successful application of IRIS methodology demonstrates its value as a technically robust and economically advantageous solution for scale-related production challenges, particularly in underperforming wells. By enabling rigless interventions, the methodology enhances well productivity, extends ESP run life, and aligns with broader environmental and operational efficiency goals, positioning IRIS as a strategic enabler for sustainable field development.

Introduction

Located in the Ecuadorian Amazon Region (Fig.1), this field has been in production since 1970 and is recognized as one of the most productive brownfields in the country. The asset is primarily produced from four sandstone formations located between 8,500 ft and 10,000 ft true vertical depth (TVD) with the deepest zone showing the highest tendency to scale formation (Rueda et al. 2023). Over the past five decades, the field has undergone extensive development and optimization efforts, including drilling, workovers, facility upgrades, and the deployment of new technologies aimed at maximizing recovery and extending production continuity (Davalos et al. 2024).



Figure 1—Remote field location map of the Ecuadorian Amazon region.

Mineral scale deposition has recently become a recurrent and disruptive flow assurance issue across the field, as a consequence of reservoir depletion, increasing water cut and changes in produced fluid composition. Although conventional mitigation practices—such as continuous chemical injection via capillary strings and periodic annular batch treatments to the wellbore— have been routinely applied, they have proven insufficient to prevent deposition both below the ESP system and across the perforated interval. As a result, a progressive restriction is generated along the flow path from the reservoir to the completion.

In 2024, a comprehensive review was conducted on all wells that showed accelerated decline trends from the beginning of their production life and had stabilized at relatively low rates. The analysis confirmed that bottomhole scaling was the primary cause of productivity loss. This diagnosis was validated during workover operations, where milling was required to remove scale deposits and allow access to the perforated interval (Fig.2). The issue currently affects more than 37% of the active wells in the field, with cumulative production losses of around 390,000 barrels of oil per year.



Figure 2—Scale deposits and laboratory characterization.

To address this issue without requiring workover rigs and full ESP retrieval, a novel solution was developed: the IRIS methodology. This approach is structured in three sequential stages. The first, Initial Recognition, focuses on identifying oil wells with a reduction in PI by analyzing ESP operating parameters, production trends, and nodal analysis. The second stage, Analysis, focuses on validating the presence of bottomhole scale by integrating production behavior, ESP performance, reservoir characteristics, fluid properties, and chemical treatment efficiency. This evaluation confirms whether the flow impairment is scale-related and supports the selection of candidates for stimulation. In the final stage, Implementation, the acid treatment is designed according to the specific conditions of each well and executed through a CTU via a sliding sleeve, enabling rigless scale removal directly at the formation face while preserving ESP integrity and minimizing deferred production.

The IRIS methodology represents a breakthrough in addressing bottomhole scaling in ESP-completed wells, offering a fully rigless solution by enabling direct acid placement at the formation face without pulling the ESP; this approach significantly reduces intervention time, minimizes production losses, optimizes resources and cost. This paper presents the first successful field application of IRIS in an amazonian brownfield, detailing the methodology, field execution, and post-treatment performance.

Statement of Theory and Definitions

Scale Deposition in Brown Fields

In fields where oil and gas have been produced for an extended period, different challenges start to come up related to production decline. Scale deposition is a well-known flow assurance issue in such brown fields, particularly in Ecuadorian assets. Scale refers to the accumulation of inorganic mineral deposits, such as calcium carbonate (CaCO_3) or barium sulfate (BaSO_4), which precipitate when the thermodynamic equilibrium of produced water is disturbed, mainly due to changes in pressure, temperature or release of dissolved gases (Neville, 2012). In some cases, the mixing of incompatible waters, typically injection and formation water can also trigger scale formation (Neville, 2012). In the context of Ecuadorian brownfields, scale challenges under ESP systems have been documented, with operational impacts linked to these thermodynamic shifts (Andagoya et al. 2022).

The formation of carbonate-based scales within the production string, artificial lift system and borehole, restricts flow paths, increases skin damage in the sand face, and creates mechanical restrictions. As reference, in the field of study, scale-related restrictions have caused up to 70% production loss in affected wells, with a direct impact on reservoir deliverability and operational continuity.

Producing high fluid volumes under low bottomhole pressure, ESP is the most suitable artificial lift system in the field. However, in wells with simple ESP completions that lack dedicated access to the wellbore for chemical intervention, conventional remediation methods become limited. This constraint

demands innovative solutions and cleaning approaches that avoid compromising the completion or incurring in the high costs and downtime associated with rig-based interventions.

A typical ESP completion consists of a bottom hole assembly (BHA) with production tubing, sliding sleeve, and a nipple, followed by the ESP assembly (Fig.3).

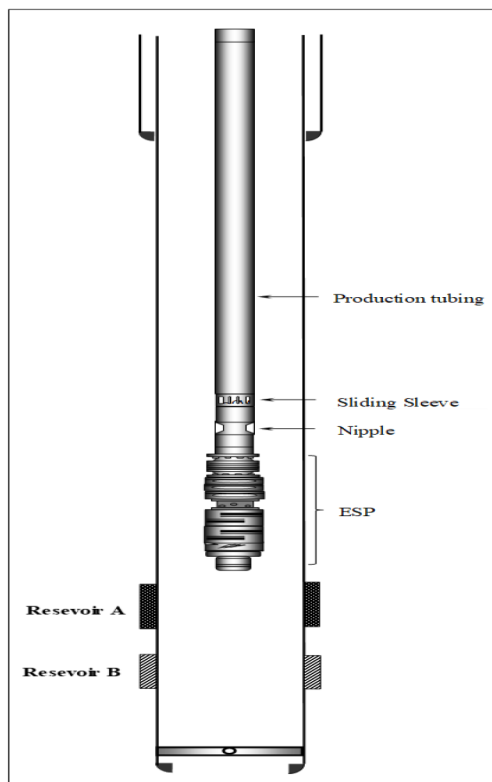


Figure 3—Well schematic with simple ESP completion.

Flow Assurance Principles

Flow assurance encompasses all strategies and technologies required to ensure uninterrupted flow from the reservoir to the surface. In this context, scale control is a critical component, especially in fields where bottomhole pressures decrease significantly with time and water cuts and mineral content increases. Operationally, uncontrolled scale precipitation may lead to:

- Reduce tubing and casing flow area.
- Increase skin damage effects at sand face.
- Increase frictional losses and decrease flowing bottomhole pressure (FBHP).
- Flow area restriction through the ESP system.
- Downhole equipment damage due to changes in wellbore hydrodynamics and chemical environments.

These effects accelerate production decline, hence the PI, and increase the frequency and cost of workover interventions. To address the mechanical limitations in ESP completions and decrease the need for a workover rig, the IRIS methodology was developed. This approach enables scale removal through rigless operations. The intervention is performed using a CTU, a surface-deployed system (Fig. 4) that allows access to the wellbore under pressure using a continuous coil, enabling chemical treatments and cleanouts without pulling out the ESP.

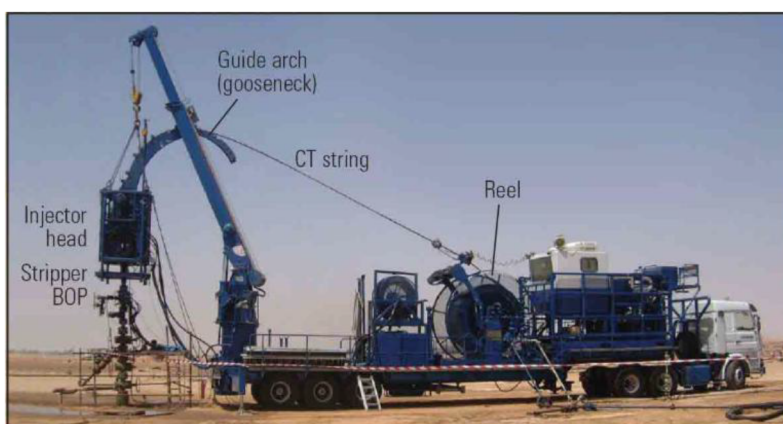


Figure 4—CTU Illustration (von Flatern, 2015). BOP = Blowout Preventer. CT = Coiled Tubing.

Productivity Index

The PI is a measure of a well's ability to produce fluids and is defined as the ratio of the production rate to the pressure drawdown. It is a primary indicator of formation performance and flow efficiency. A declining PI may suggest formation damage, near-wellbore plugging (e.g., scale, fines), or mechanical restrictions, and is commonly used in diagnostic workflows for artificial lift wells (Economides et al. 2000; Lee et al. 2019).

Nodal Analysis. Nodal analysis is a system approach to evaluate wells' performance by analyzing pressure and flow relationships throughout the production system. This technique enables the estimation of well potential, flow restrictions, and production optimization. A shift in the operating point or a mismatch between measured and predicted rates can signal issues like scale buildup, gas interference, or equipment malfunction (Brown, 1984; Lea et al. 2008). Nodal analysis plays a crucial role in the methodology described in this paper, particularly for identifying reductions in well PI and evaluating the impact of wellbore restrictions below the ESP system.

Description and Application of Innovative Methodology and Process

The IRIS workflow was developed progressively through field experience, evolving from the initial interventions performed on a limited number of wells. These early applications served as the foundation for refining both diagnostic criteria and operational procedures, enabling a more robust and structured methodology. As more wells were treated, key engineering learnings were incorporated to enhance candidate selection, reduce operational uncertainty, and standardize decision-making. The methodology was ultimately consolidated into three sequential stages—Initial Recognition, Analysis, and Implementation, which together provide a comprehensive framework for identifying, validating, and executing rigless acid stimulations in ESP-completed wells affected by bottomhole scaling (Fig. 5).

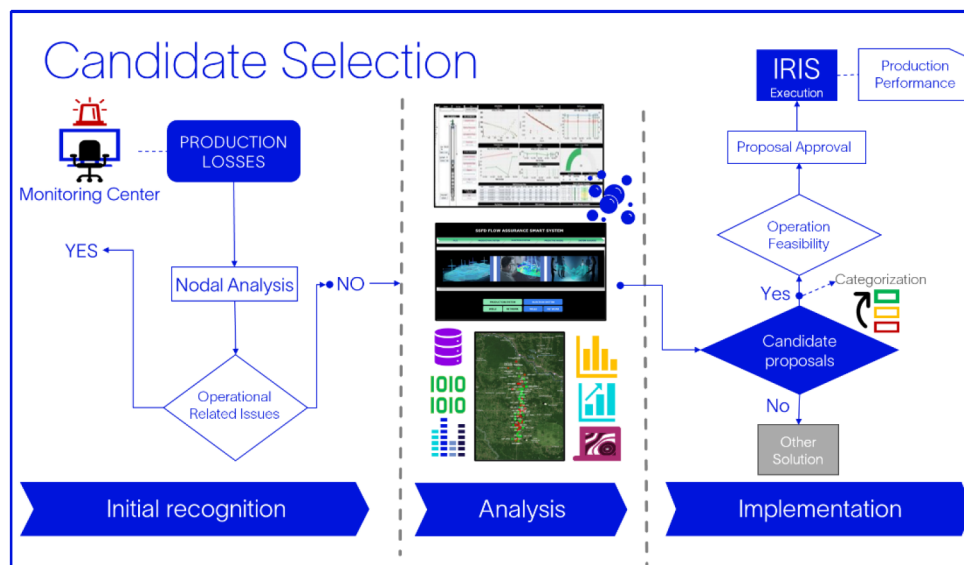


Figure 5—Candidate selection workflow.

Stage 1: Initial Recognition

The workflow begins by identifying wells exhibiting unexpected behavior in their production performance, based on real-time monitoring of ESP operating parameters. Automated surveillance systems flag deviations such as changes in pump intake pressure (PIP), motor current, pump discharge pressure (PDP), and fluid load, which may indicate flow restriction or downhole anomalies.

The Integrated Asset Management (MIA) Center is a surveillance department responsible for monitoring production and ESP performance parameters. When a production loss is detected, the center triggers an alert and engineering validation is performed to determine whether the anomaly is potentially related to a reduction in the well PI or due to operational issues. This validation includes updating the well test and the nodal analysis to evaluate the inflow and outflow balance. Key indicators such as reduced drawdown response, mismatch between expected and actual flow rates, and deviations in pump efficiency are used to support the diagnosis. If the behavior is confirmed to be associated with a decline in PI—particularly in the absence of mechanical failures or surface constraints—the well is flagged as a potential candidate for bottomhole stimulation under the IRIS methodology. On the other hand, alerts unrelated to PI reduction, such as electrical issues, surface facility restrictions, sensor drift, gas interference, tubing leaks, or solids handling problems, are flagged as operational related issues and addressed through alternative pathways outside the scope of this methodology.

This structured recognition stage ensures that only wells with a high likelihood of wellbore-to-production casing flow restriction—typically due to bottomhole scaling—proceed to the next phase of analysis and treatment design, thereby optimizing resource allocation and minimizing unnecessary rig interventions.

Stage 2: Analysis

The second stage of the IRIS methodology focuses on an exhaustive diagnostic process to validate the presence of bottomhole scale and reduce uncertainty prior to intervention. This phase integrates a wide range of multidisciplinary data sources to contrast multiple hypotheses and strengthen the technical basis for candidate selection.

Key datasets include ESP operating trends, production test history, mechanical well conditions, surface constraints, reservoir properties, petrophysical interpretation, fluid characterization, daily operational performance metrics of applied chemical treatment, surface conditions, past workover and historical intervention records. These elements are analyzed in parallel to build a consistent diagnostic framework.

Production decline is compared against ESP performance metrics to distinguish between reservoir-driven depletion and flow restriction scenarios, validate the results from stage 1 and integrate them into additional analysis. Dynamic parameters such as flowing pressure and production evolution are complemented with petrophysical information and compared with offset wells to determine whether the observed behavior is consistent across the area, or specific to the well under evaluation. Simultaneously, chemical treatment efficiency and data trends are reviewed to detect changes in scale risk indicators.

This multi-criteria analysis, supported by evidence gathered during the Initial Recognition phase and guided by engineering judgment, aims to build a consistent diagnostic case for scale-related flow restriction. Once the data indicates a high probability of scale formation around the perforated interval—and other alternative causes have been reasonably excluded—the well is classified as a suitable candidate for rigless stimulation under the IRIS methodology. Additionally, for each candidate, an economic evaluation is performed based on the estimated incremental oil production and the well intervention costs to confirm the overall feasibility of the treatment.

Stage 3: Implementation

Based on the diagnostic results from the Analysis phase, a rigless-intervention technical proposal is developed for each candidate well. It includes defining the treatment design according to specific conditions, selecting the type and volume of fluids, acid concentration, additives, and exposure time required to effectively dissolve scale deposits. Wells are then classified according to risk and expected success (Table 1).

Table 1—Candidate category.

Risk Success Probability	Low	Moderate	High
Low	-	-	C
Moderate	-	B	-
High	A	-	-

Once the wells are categorized, the results provide insights to guide decision-making, as follows:

- A. Low risk / high success probability → immediate candidate
- B. Moderate risk / moderate probability → potential future candidate
- C. High risk / low probability → not eligible for intervention

Category "A" wells are immediate candidates, since they are prioritized for intervention; an operational feasibility assessment is performed, and necessary approvals are managed in parallel to ensure minimal operational delay. Once validated and authorized, the intervention is executed using a fully rigless approach.

Category "B" wells are candidates for intervention; however, the current production loss does not justify immediate action. They undergo continuous monitoring until the associated incremental production justifies the operational risk, considering that its production loss may still be in an early stage.

Category "C" wells candidates are not eligible for rigless intervention due to operational risks or low incremental production expected. Therefore, they are scheduled for workover intervention, during which complementary operations can be executed to ensure full production optimization.

The mentioned rigless operation is executed through a structured sequence of activities, described as follows: the well is shut in, a slickline unit is deployed to open the sliding sleeve installed in the production string and a standing valve is installed in the no-go nipple if needed. Subsequently, a CT is run in hole to the sliding sleeve depth to pump control fluid between the annulus and the ESP to perform a reservoir admission test. If the admission is confirmed, the chemical treatment is pumped and displaced under overbalanced conditions to ensure effective placement across the perforated intervals targeting both sand face stimulation

and scale rings removal beneath the ESP. After treatment, the CT is retrieved, the slick line unit closes the sliding sleeve, and the well starts up.

Post-stimulation evaluation involves monitoring ESP parameters and updating tests to confirm recovery of the well's PI (Fig. 6). Lessons learned are documented and shared with all operational teams involved to promote continuous improvement and support future interventions under the IRIS methodology.

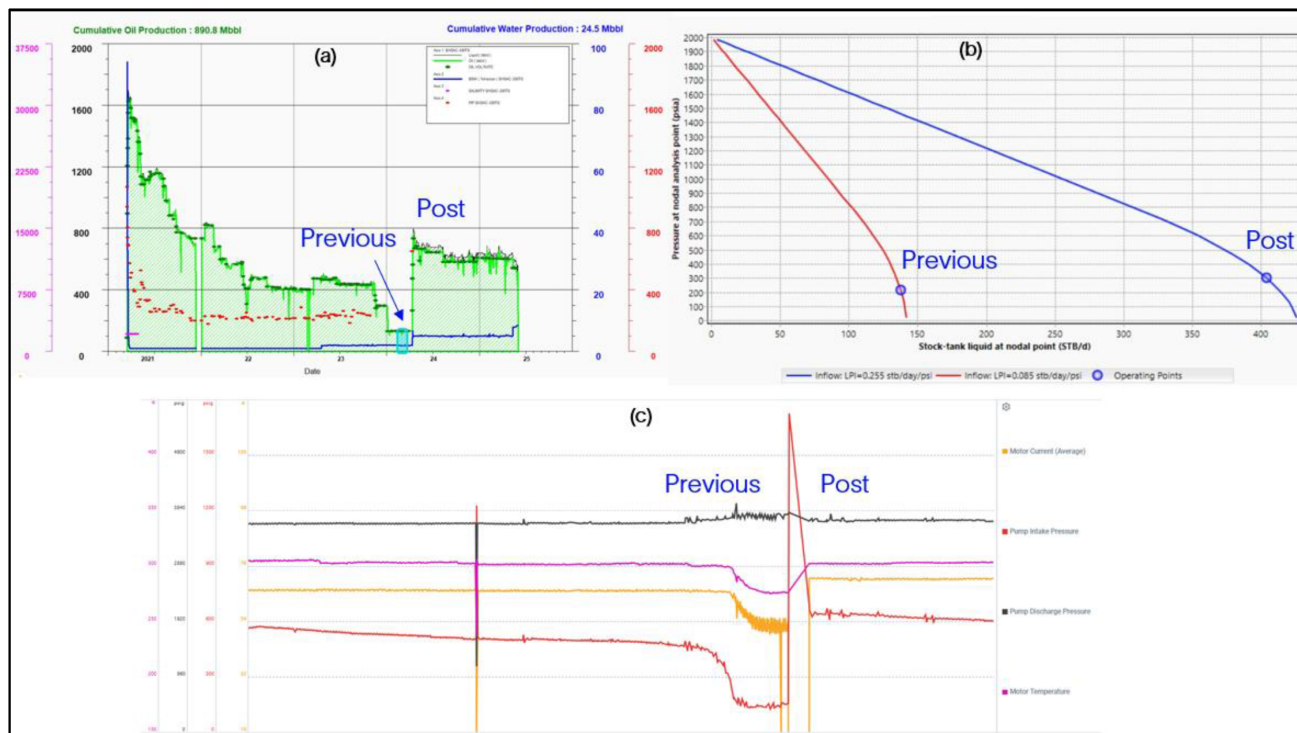


Figure 6—Previous and post IRIS intervention. (a) Historical production, (b) Productivity index, (c) ESP parameters.

Results

The IRIS methodology was applied in 18 simple completion wells with "A" categorization, that were showing a sharp decline in PIP, reduction in their PI, and incrusting tendency. The implementation of this methodology during 2024 has yielded outstanding results by effectively eliminating the scale at sand face, improving the PI by 200% – 300%, and achieving a cumulative production increase of 1,060 BOPD year average representing \$25 million USD in revenue. Compared to conventional methods using a rig, the well intervention time was reduced from 15 to 2 days, and its downtime and production losses during intervention decreased by 75 %. In addition, operational costs per well were cut by 85%.

On the other hand, field personnel trips decreased by 55% and operations emissions were lowered at least 75%, avoiding 500-t CO₂ equivalent emissions per year (Fig. 7). Furthermore, the probability of Health, Safety, and Environment (HSE) incidents during intervention was significantly avoided. These results demonstrate that IRIS not only optimizes production and reduces costs, but also enhances safety and sustainability, establishing it as a comprehensive solution for rigless well interventions.

This methodology extends the productive life of both the well and the ESP completion by eliminating restrictions caused by scale deposition at sand face, particularly in wells where conventional CT access is not possible. By restoring flow in otherwise underperforming wells, the intervention gives the well a second chance to produce efficiently, without the need for complex and high-cost workovers. It also improves ESP run life, supporting system integrity and operational continuity while aligning with the asset field development plan and sustainability goals.

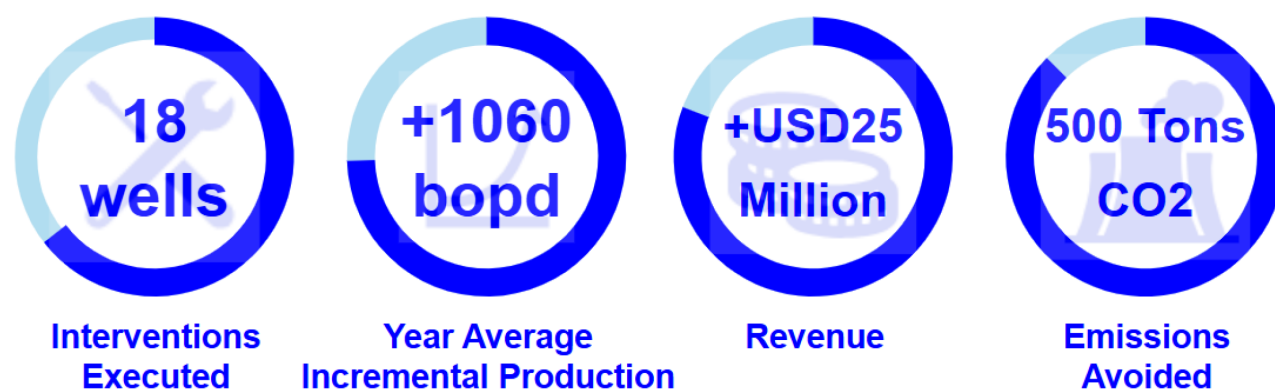


Figure 7—IRIS methodology results.

The absence of a rig simplifies logistics, reduces operational complexity, and shortens planning and mobilization timelines. These interventions are less intrusive, enabling quicker response to production issues and reducing operational downtime, which is critical in maintaining reservoir connectivity and asset performance. Rigless operations also enhance scheduling flexibility, contractual agreements, and streamline coordination between the operator and service companies.

Conclusions

The implementation of the IRIS methodology has proven to be a highly effective solution for tackling production challenges associated with scale formation in wells with sensitive completions or limited intervention capabilities. In brown fields with severe scaling issues, this technique successfully restored productivity without the need to pull out the completion or mobilize a workover rig, eliminating a major operational constraint in ESP simple wells. Its rigless, simple, and highly targeted nature makes it a strategic tool for extending the productive life of wells, maintaining operational continuity, and minimizing the environmental footprint of wells interventions. By eliminating the rig and compressing operational timeframes, this methodology drastically reduces emissions, resource consumption and waste, while minimizing local environmental and social disturbances.

Economically, the methodological approach generated \$25 million USD in combined production gains and cost savings, while improving equipment run life and asset uptime. The technique has also enabled operational continuity in wells that would otherwise be constrained by completion limitations, particularly those affected by scale. Rigless interventions represent a scalable, efficient, and sustainable alternative to conventional workovers.

These results establish IRIS as a technically and economically superior alternative to conventional interventions methods for wells where this methodology could be applied. It not only improves productivity but also supports field development plans by combining technical efficacy with measurable environmental and social responsibility. The methodology is especially suited for fields with constrained wells, where prompt, effective, and non-complex interventions are critical to maintain reservoir deliverability and minimize downtime.

Acknowledgments

The authors gratefully acknowledge the operational team of the National Oil Company for granting access to field data and inputs, as well as for providing valuable feedback, which contributed to the successful application of the methodology presented for rigless intervention. The authors further acknowledge the support and collaboration of all the stakeholder's teams throughout the execution of this work.

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